PDEs Problem Sheet 6

1. The function $V(x,y)$ satisfies Laplace's equation

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0$$

in the region R.

   (a) Show that, when R is the region $y > 0$, $0 < x < a$, then

$$V(x,y) = Ae^{-n\pi y/a} \sin\left(\frac{n\pi x}{a}\right),$$

   where n is an integer and A is constant, satisfies Laplace's equation and the boundary conditions

$$V \to 0 \quad \text{as} \quad y \to \infty,$$
$$V = 0 \quad \text{at} \quad x = 0 \quad \text{and} \quad x = a \quad \text{for} \quad y > 0.$$

   Find A and n such that

$$V(x,0) = 100 \sin\left(\frac{2\pi x}{a}\right) \quad \text{for} \quad 0 < x < a.$$

   (b) Show that, when R is the region $0 \le x \le a$, $0 \le y \le b$, then

$$V(x,y) = A + B \sinh\left(\frac{n\pi y}{a}\right) \cos\left(\frac{n\pi x}{a}\right),$$

   where n is an integer and A is constant, satisfies Laplace's equation and the boundary conditions

$$\frac{\partial V}{\partial x} = 0 \quad \text{at} \quad x = 0 \text{ and } x = a \quad \text{for} \quad 0 < y < b.$$

   Find A, B and n such that

$$V(x,0) = 0 \quad \text{and} \quad V(x,b) = \cos\left(\frac{\pi x}{a}\right) \quad \text{for} \quad 0 < x < a.$$

2. The function $T(r,\theta)$ satisfies Laplace's equation in polar coordinates (r,θ)

$$\frac{\partial^2 T}{\partial r^2} + \frac{1}{r}\frac{\partial T}{\partial r} + \frac{1}{r^2}\frac{\partial^2 T}{\partial \theta^2} = 0$$

in the region R.

Show that if R is the region $a < r < b$, $0 \le \theta < 2\pi$, then

$$T = A + B \ln r,$$

where A and B are constants, satisfies Laplace's equation and find A and B such that

$$T(a,\theta) = T_1 \quad \text{and} \quad T(b,\theta) = T_2,$$

where T_1 and T_2 are given constants.

3. The function $U(x,t)$ satisfies the one-dimensional heat conduction (diffusion) equation

$$\frac{\partial^2 U}{\partial x^2} - \frac{1}{K}\frac{\partial U}{\partial t} = 0,$$

where K is the thermal diffusivity in a bar $0 < x < a$.
Show that

$$U(x,t) = A \sin\left(\frac{n\pi x}{a}\right) \exp\left(-\frac{n^2\pi^2 K t}{a^2}\right),$$

where n is an integer and A is constant, satisfies the heat equation and the boundary conditions

$$U(0,t) = 0, \quad U(a,t) = 0 \quad \text{for} \quad t > 0.$$

Find A and n such that

$$U(x,0) = 10 \sin\left(\frac{\pi x}{a}\right).$$

Prerequisites:
$P(n) = \{$questions with prerequisite Parts $\leq n\}$
$A =$ all of PS6.

$P(15) = P(16) = A$ (whilst Part 15 focusses on the Wave Equation, all questions should be accessible following Part 15)